

Artificial Artists: Can Computers Really Create Works of Art?

From robot painters to AI composers — exploring the frontier where creativity and code collide.

computational creativity · creative process · irresistible essence · pseudoscience · enraptured
· ghostlike · prejudice · indistinguishable · composition · artificial intelligence

A The Painting Fool is one of a growing number of computer programs which, so their makers claim, possess creative talents. Classical music by an artificial composer has had audiences enraptured, and even tricked them into believing a human was behind the score. Artworks painted by a robot have sold for thousands of dollars and been hung in prestigious galleries. And software has been built which creates art that could not have been imagined by the programmer.

B Human beings are the only species to perform sophisticated creative acts regularly. If we can break this process down into computer code, where does that leave human creativity? 'This is a question at the very core of humanity,' says Geraint Wiggins, a computational creativity researcher at Goldsmiths, University of London. 'It scares a lot of people. They are worried that it is taking something special away from what it means to be human.'

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C To some extent, we are all familiar with computerised art. The question is: where does the work of the artist stop and the creativity of the computer begin? Aaron is one of the oldest machine artists — a robot that has had paintings exhibited in London's Tate Modern and the San Francisco Museum of Modern Art. Aaron can pick up a paintbrush and paint on canvas on its own. Impressive perhaps, but it is still little more than a tool to realise the programmer's own creative ideas.

D Simon Colton, the designer of the Painting Fool, is keen to make sure his creation doesn't attract the same criticism. Unlike earlier 'artists' such as Aaron, the Painting Fool only needs minimal direction and can come up with its own concepts by going online for material. The software runs its own web searches and trawls through social media sites. It is now beginning to display a kind of imagination too, creating pictures from scratch. One of its original works is a series of fuzzy landscapes, depicting trees and sky. While some might say they have a mechanical look, Colton argues that such reactions arise from people's double standards towards software-produced and human-produced art. After all, he says, consider that the Painting Fool painted the landscapes without referring to a photo. 'If a child painted a new scene from its head, you'd say it has a certain level of imagination,' he points out. 'The same should be true of a machine.' Software bugs can also lead to unexpected results. Some of the Painting Fool's paintings of a chair came out in black and white, thanks to a technical glitch. This gives the work an eerie, ghostlike quality. Human artists like the renowned Ellsworth Kelly are lauded for limiting their colour palette — so why should computers be any different?

PART TWO · THE DEBATE OVER MACHINE CREATIVITY

E Researchers like Colton don't believe it is right to measure machine creativity directly to that of humans who 'have had millennia to develop our skills'. Others, though, are fascinated by the prospect that a computer might create something as original and subtle as our best artists. So far, only one has come close. Composer David Cope invented a program called Experiments in Musical Intelligence, or EMI. Not only did EMI create compositions in Cope's style, but also that of the most revered classical composers, including Bach, Chopin and Mozart. Audiences were moved to tears, and EMI even fooled classical music experts into thinking they were hearing genuine Bach. Not everyone was impressed however. Some, such as Wiggins, have blasted Cope's work as pseudoscience, and condemned him for his deliberately vague explanation of how the software worked. Meanwhile, Douglas Hofstadter of Indiana University said EMI created replicas which still rely completely on the original artist's creative impulses. When audiences found out the truth they were often outraged with Cope, and one music lover even tried to punch him. Amid such controversy, Cope destroyed EMI's vital databases.

■ The Moffat Study — Glasgow Caledonian University

But why did so many people love the music, yet recoil when they discovered how it was composed? A study by computer scientist David Moffat of Glasgow Caledonian University provides a clue. He asked both expert musicians and non-experts to assess six compositions. The participants weren't told beforehand whether the tunes were composed by humans or computers, but were asked to guess, and then rate how much they liked each one. People who thought the composer was a computer tended to dislike the piece more than those who believed it was human. This was true even among the experts, who might have been expected to be more objective in their analyses.

F Where does this **prejudice** come from? Paul Bloom of Yale University has a suggestion: he reckons part of the pleasure we get from art stems from the **creative process** behind the work. This can give it an '**irresistible essence**', says Bloom. Meanwhile, experiments by Justin Kruger of New York University have shown that people's enjoyment of an artwork increases if they think more time and effort was needed to create it. Similarly, Colton thinks that when people experience art, they wonder what the artist might have been thinking or what the artist is trying to tell them. It seems obvious, therefore, that with computers producing art, this speculation is cut short — there's nothing to explore. But as technology becomes increasingly complex, finding those greater depths in computer art could become possible. This is precisely why Colton asks the Painting Fool to tap into online social networks for its inspiration: hopefully this way it will choose themes that will already be meaningful to us.

■ TALKING POINTS

Wiggins says computer creativity 'scares a lot of people'. Why do you think human creativity feels so fundamental to our sense of identity?

Do you think Aaron and the Painting Fool represent genuine creativity — or are they just very sophisticated tools? Where is the line?

Audiences felt betrayed when they discovered EMI's music was computer-generated. Do you think this reaction is rational or unfair?

Bloom argues that art gains its value partly from the creative process behind it. If you could not tell the difference, would computer art have the same value as human art?

Name: _____

Date: _____

Score: _____ / 14

LEVEL B • WORKSHEET

Worksheet: Artificial Artists

IELTS-style comprehension tasks — Questions 27–40

PART A — MULTIPLE CHOICE [5 marks] Questions 27–31

Choose the correct letter, **A**, **B**, **C** or **D**.

Write the correct letter in the box next to each question.

27 What is the writer suggesting about computer-produced works in the first paragraph?

- A** People's acceptance of them can vary considerably.
- B** A great deal of progress has already been attained in this field.
- C** They have had more success in some artistic genres than in others.
- D** The advances are not as significant as the public believes them to be.

Answer: _____

28 According to Geraint Wiggins, why are many people worried by computer art?

- A** It is aesthetically inferior to human art.
- B** It may ultimately supersede human art.
- C** It undermines a fundamental human quality.
- D** It will lead to a deterioration in human ability.

Answer: _____

29 What is a key difference between Aaron and the Painting Fool?

- A** its programmer's background
- B** public response to its work
- C** the source of its subject matter
- D** the technical standard of its output

Answer: _____

30 What point does Simon Colton make in the fourth paragraph?

- A** Software-produced art is often dismissed as childish and simplistic.
- B** The same concepts of creativity should not be applied to all forms of art.
- C** It is unreasonable to expect a machine to be as imaginative as a human being.
- D** People tend to judge computer art and human art according to different criteria.

Answer: _____

31 **The writer refers to the paintings of a chair as an example of computer art which**

A achieves a particularly striking effect.

B exhibits a certain level of genuine artistic skill.

C closely resembles that of a well-known artist.

D highlights the technical limitations of the software.

Answer: _____

PART B – SENTENCE ENDINGS [6 marks] Questions 32–37

Complete each sentence with the correct ending, **A–G** below.

Write the correct letter, A–G, in the box next to each question.

List of Ideas

A generating work that was virtually indistinguishable from that of humans.

B knowing whether it was the work of humans or software.

C producing work entirely dependent on the imagination of its creator.

D comparing the artistic achievements of humans and computers.

E revealing the technical details of his program.

F persuading the public to appreciate computer art.

G discovering that it was the product of a computer program.

32 Simon Colton says it is important to consider the long-term view when

Answer: _____

33 David Cope's EMI software surprised people by

Answer: _____

34 Geraint Wiggins criticised Cope for not

Answer: _____

35 Douglas Hofstadter claimed that EMI was

Answer: _____

36 Audiences who had listened to EMI's music became angry after

Answer: _____

37

The participants in David Moffat's study had to assess music without

Answer: _____

PART C — NOTE COMPLETION [3 marks] Questions 38–40

Complete the notes below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Write your answers in the boxes below.

Why people respond negatively to computer-generated art • According to Paul Bloom, part of the pleasure of art comes from the **38** _____ behind the work, which he calls its 'irresistible essence'. • Justin Kruger's research shows that enjoyment increases when people believe more **39** _____ was needed to produce the artwork. • Simon Colton argues that people feel that with computer art, **40** _____ is cut short — there is nothing to explore.

38.

39.

40.

ANSWER KEY

Artificial Artists — Answer Key

Teacher's copy — do not distribute to students

Part	Task Type	Questions	Marks
A	Multiple Choice (A/B/C/D)	27–31	5
B	Sentence Endings (A–G)	32–37	6
C	Note Completion (max 2 words)	38–40	3
TOTAL			14

Questions 27–31 — Multiple Choice

Accept only the single correct letter. No partial marks.

27. **B** Para A lists multiple impressive achievements: classical music, robot paintings, and software creating unimagined art — showing the breadth of progress already made.
28. **C** Wiggins says people worry it is 'taking something special away from what it means to be human' — undermining a fundamental quality of human identity.
29. **C** Aaron works from the programmer's ideas; the *Painting Fool* goes online and finds its own material — the source of subject matter is the key difference.
30. **D** Colton argues people apply double standards: they criticise computer art for things (limited palette, painting from imagination) they praise in human artists.
31. **C** The chair paintings, due to a glitch, had the same limited palette as Ellsworth Kelly's work — the passage explicitly draws this comparison.

Questions 32–37 — Sentence Endings

Accept only the single correct letter. No partial marks.

32. **D** Colton says machines shouldn't be compared directly to humans 'who have had millennia to develop our skills' — he wants the long-term view when comparing artistic achievements.
33. **A** EMI fooled classical music experts into thinking they were hearing genuine Bach — virtually indistinguishable from human composition.
34. **E** Wiggins condemned Cope for his 'deliberately vague explanation of how the software worked' — not revealing the technical details.

35. **C** *Hofstadter said EMI created 'replicas which still rely completely on the original artist's creative impulses' — entirely dependent on the original creator.*
36. **G** *'When audiences found out the truth they were often outraged' — angry after discovering the music was a computer program's product.*
37. **B** *The participants 'weren't told beforehand whether the tunes were composed by humans or computers' — they had to assess without knowing the source.*

Questions 38–40 — Note Completion

Accept only words taken directly from the passage. Maximum TWO WORDS per answer.

38. **creative process** *Para F: 'part of the pleasure we get from art stems from the creative process behind the work.'*
39. **time and effort** *Para F: 'people's enjoyment of an artwork increases if they think more time and effort was needed to create it.'*
40. **this speculation** *Para F: 'with computers producing art, this speculation is cut short — there's nothing to explore.'*

MARKING SUMMARY

Part A (Multiple choice Qs 27–31): 5 marks — 1 per question · Part B (Sentence endings Qs 32–37): 6 marks — 1 per question · Part C (Note completion Qs 38–40): 3 marks — 1 per question · TOTAL: 14 marks